Proposal: Fuzzing-Guided Failure Discovery and Selective Recovery in ALFWorld

We study failure propagation and recovery in a multi-agent ALFWorld workflow. Agents produce intermediate artifacts such as observations, memory entries, and plans; faults in these artifacts may propagate into downstream reasoning, actions, and environment state.

Method

We build a lightweight multi-agent workflow on top of ALFWorld with an observation/state agent, memory agent, planner, and executor. Each intermediate result is recorded in a dependency graph.

In **Phase 1**, a fuzzing-inspired fault injector mutates intermediate artifacts using incorrect, stale, missing, or contradictory information. Instead of keeping random mutations, the fuzzer prioritizes cases that produce new behaviors, propagation paths, or task failures.

In **Phase 2**, once a faulty artifact is identified, we compare three recovery strategies:

1. **Full restart** — rerun the entire mission.
2. **Dependency-based recovery** — recompute all downstream dependents.
3. **Influence-aware recovery** — recompute incrementally and stop when correcting the fault no longer changes downstream outputs.

ALFWorld state is recovered using reset-and-replay: valid earlier actions are replayed, while agent reasoning is recomputed only from the first affected point.

Research Questions

- Can fuzzing discover more diverse failure-propagation cases than random fault injection?
- How do faults propagate through agent reasoning and environment state?
- Can selective recovery reduce recomputation while preserving task correctness?

Evaluation

We measure task success, failure diversity, propagation depth, recomputed agent calls, work preserved, recovery latency, API cost, and tracking overhead.

Goal: combine automated failure discovery with dependency-aware selective recovery to understand and efficiently repair failures in multi-agent systems.